

New Xist-Interacting Proteins in X-Chromosome Inactivation.

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To achieve dosage compensation of X-linked gene expression, female mammalian cells inactivate one X chromosome through a process called X-chromosome inactivation (XCI). A central component of this process is the X-encoded long non-coding RNA Xist. Following upregulation from one X chromosome, Xist spreads in cis, kicking off a plethora of events that ultimately results in stable X-linked gene repression, which is then faithfully transmitted to all daughter cells. In the last decades, intensive work has been undertaken to understand each of the steps in XCI, namely Xist transcription control, Xist spreading and localization, and silencing of gene expression. Recently, several groups have spearheaded the research of Xist's interactome and the factors involved in silencing. Several novel proteins have now been shown to be required for the transcriptional silencing of the X chromosome and/or Xist spreading and localization to the inactive X chromosome. Here, we review these new findings in the context of existing knowledge about Xist-interacting factors.

Evidence that heteronuclear proteins interact with XIST RNA in vitro.

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The process of X chromosome inactivation results in the transcriptional silencing of one of the two X chromosomes in mammalian females. A large heterogeneous nuclear RNA that is expressed exclusively from the inactive X chromosome (XIST--X Inactive Specific Transcripts) has been implicated in the inactivation process. The XIST RNA colocalizes with the inactive X chromosome and therefore proteins that interact with the XIST RNA may be involved in the inactivation of the X chromosome. In order to identify such proteins we have used an in vitro UV light cross-linking technique to detect nuclear proteins associating with sections of the XIST RNA. The strongest interaction detected by this technique was between a pair of approximately 40 kDa proteins and a 5' region of the XIST RNA which contains a series of well-conserved tandem repeats. Immunoprecipitation suggested that these proteins may be the heteronuclear proteins hnRNP1/C2.

Chromosomal silencing and localization are mediated by different domains of Xist RNA.

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The gene Xist initiates the chromosomal silencing process of X inactivation in mammals. Its product, a noncoding RNA, is expressed from and specifically associates with the inactive X chromosome in female cells. Here we use an inducible Xist expression system in mouse embryonic stem cells that recapitulates long-range chromosomal silencing to elucidate which Xist RNA sequences are necessary for chromosomal association and silencing. We show that chromosomal association and spreading of Xist RNA can be functionally separated from silencing by specific mutations. Silencing requires a conserved repeat sequence located at the 5' end of Xist. Deletion of this element results in Xist RNA that still associates with chromatin and spreads over the chromosome but does not effect transcriptional repression. Association of Xist RNA with chromatin is mediated by functionally redundant sequences that act cooperatively and are dispersed throughout the remainder of Xist but show little or no homology.

Xist localization and function: new insights from multiple levels.

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In female mammals, one of the two X chromosomes in each cell is transcriptionally silenced in order to achieve dosage compensation between the genders in a process called X chromosome inactivation. The master regulator of this process is the long non-coding RNA Xist. During X-inactivation, Xist accumulates in cis on the future inactive X chromosome, triggering a cascade of events that provoke the stable silencing of the entire chromosome, with relatively few genes remaining active. How Xist spreads, what are its binding sites, how it recruits silencing factors and how it induces a specific topological and nuclear organization of the chromatin all remain largely unanswered questions. Recent studies have improved our understanding of Xist localization and the proteins with which it interacts, allowing a reappraisal of ideas about Xist function. We discuss recent advances in our knowledge of Xist-mediated silencing, focusing on Xist spreading, the nuclear organization of the inactive X chromosome, recruitment of the polycomb complex and the role of the nuclear matrix in the process of X chromosome inactivation.

A protein assembly mediates Xist localization and gene silencing.

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Nuclear compartments have diverse roles in regulating gene expression, yet the molecular forces and components that drive compartment formation remain largely unclear¹. The long non-coding RNA Xist establishes an intra-chromosomal compartment by localizing at a high concentration in a territory spatially close to its transcription locus² and binding diverse proteins³⁻⁵ to achieve X-chromosome inactivation (XCI)^{6,7}. The XCI process therefore serves as a paradigm for understanding how RNA-mediated recruitment of various proteins induces a functional compartment. The properties of the inactive X (Xi)-compartment are known to change over time, because after initial Xist spreading and transcriptional shutoff a state is reached in which gene silencing remains stable even if Xist is turned off⁸. Here we show that the Xist RNA-binding proteins PTBP19, MATR310, TDP-4311 and CELF112 assemble on the multivalent E-repeat element of Xist⁷ and, via self-aggregation and heterotypic protein-protein interactions, form a condensate¹ in the Xi. This condensate is required for gene silencing and for the anchoring of Xist to the Xi territory, and can be sustained in the absence of Xist. Notably, these E-repeat-binding proteins become essential coincident with transition to the Xist-independent XCI phase⁸, indicating that the condensate seeded by the E-repeat underlies the developmental switch from Xist-dependence to Xist-independence. Taken together, our data show that Xist forms the Xi compartment by seeding a heteromeric condensate that consists of ubiquitous RNA-binding proteins, revealing an unanticipated mechanism for heritable gene silencing.

Identification of RNA-Protein Interactions Through In Vitro RNA Pull-Down Assays.

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Recent advances in next-generation sequencing have revealed that majority of the human genome is transcribed into long and short RNA (ncRNA) transcripts. Many ncRNAs function by interacting with proteins and forming regulatory complexes. RNA-protein interactions are vital in controlling core cellular processes like transcription and translation. Therefore identifying proteins that interact with ncRNAs is

central to deciphering ncRNA functions. Here we describe an RNA-protein pull-down assay, which enables the identification of proteins that interact with an RNA under study. As an example we describe pull-down of proteins interacting with ncRNA XIST, which assists in the recruitment of the polycomb-repressive complex-2 (PRC2) and drives X-chromosomal inactivation.

XIST loss impairs mammary stem cell differentiation and increases tumorigenicity through Mediator hyperactivation.

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X inactivation (XCI) is triggered by upregulation of XIST, which coats the chromosome in cis, promoting formation of a heterochromatic domain (Xi). XIST role beyond initiation of XCI is only beginning to be elucidated. Here, we demonstrate that XIST loss impairs differentiation of human mammary stem cells (MaSCs) and promotes emergence of highly tumorigenic and metastatic carcinomas. On the Xi, XIST deficiency triggers epigenetic changes and reactivation of genes overlapping Polycomb domains, including Mediator subunit MED14. MED14 overdosage results in increased Mediator levels and hyperactivation of the MaSC enhancer landscape and transcriptional program, making differentiation less favorable. We further demonstrate that loss of XIST and Xi transcriptional instability is common among human breast tumors of poor prognosis. We conclude that XIST is a gatekeeper of human mammary epithelium homeostasis, thus unveiling a paradigm in the control of somatic cell identity with potential consequences for our understanding of gender-specific malignancies.

Xist recruits the X chromosome to the nuclear lamina to enable chromosome-wide silencing.

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The Xist long noncoding RNA orchestrates X chromosome inactivation, a process that entails chromosome-wide silencing and remodeling of the three-dimensional (3D) structure of the X chromosome. Yet, it remains unclear whether these changes in nuclear structure are mediated by Xist and whether they are required for silencing. Here, we show that Xist directly interacts with the Lamin B receptor, an integral component of the nuclear lamina, and that this interaction is required for Xist-mediated silencing by recruiting the inactive X to the nuclear lamina and by doing so enables Xist to spread to actively transcribed genes across the X. Our results demonstrate that lamina recruitment changes the 3D structure of DNA, enabling Xist and its silencing proteins to spread across the X to silence transcription.

Multifaceted role of CTCF in X-chromosome inactivation.

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Therian female mammals compensate for the dosage of X-linked gene expression by inactivating one of the X-chromosomes. X-inactivation is facilitated by the master regulator Xist long non-coding RNA, which coats the inactive-X and facilitates heterochromatinization through recruiting different chromatin

modifiers and changing the X-chromosome 3D conformation. However, many mechanistic aspects behind the X-inactivation process remain poorly understood. Among the many contributing players, CTCF has emerged as one of the key players in orchestrating various aspects related to X-chromosome inactivation by interacting with several other protein and RNA partners. In general, CTCF is a well-known architectural protein, which plays an important role in chromatin organization and transcriptional regulation. Here, we provide significant insight into the role of CTCF in orchestrating X-chromosome inactivation and highlight future perspectives.

SPEN integrates transcriptional and epigenetic control of X-inactivation.

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Xist represents a paradigm for the function of long non-coding RNA in epigenetic regulation, although how it mediates X-chromosome inactivation (XCI) remains largely unexplained. Several proteins that bind to Xist RNA have recently been identified, including the transcriptional repressor SPEN1-3, the loss of which has been associated with deficient XCI at multiple loci²⁻⁶. Here we show in mice that SPEN is a key orchestrator of XCI in vivo and we elucidate its mechanism of action. We show that SPEN is essential for initiating gene silencing on the X chromosome in preimplantation mouse embryos and in embryonic stem cells. SPEN is dispensable for maintenance of XCI in neural progenitors, although it significantly decreases the expression of genes that escape XCI. We show that SPEN is immediately recruited to the X chromosome upon the upregulation of Xist, and is targeted to enhancers and promoters of active genes. SPEN rapidly disengages from chromatin upon gene silencing, suggesting that active transcription is required to tether SPEN to chromatin. We define the SPOC domain as a major effector of the gene-silencing function of SPEN, and show that tethering SPOC to Xist RNA is sufficient to mediate gene silencing. We identify the protein partners of SPOC, including NCoR/SMRT, the m6A RNA methylation machinery, the NuRD complex, RNA polymerase II and factors involved in the regulation of transcription initiation and elongation. We propose that SPEN acts as a molecular integrator for the initiation of XCI, bridging Xist RNA with the transcription machinery-as well as with nucleosome remodellers and histone deacetylases-at active enhancers and promoters.
